Instructor Solution

Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Fall 2025

Instructor – Dr. Aamir Hasan - Section L2

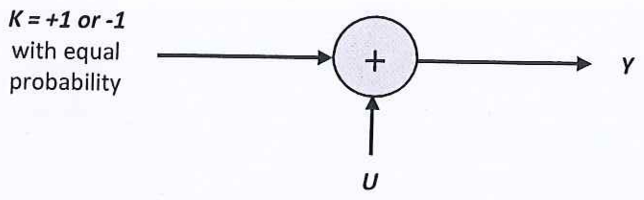
Time = 20 Minutes **Quiz No 10** [20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

Question 1 [20 points] Bayes Rule

Consider the following system comprising of three Random Variables,

- K (input & unobserved whose prior probabilities are known),
- U (Addition of random Noise),
- Y (output and the observed).



Part 1 [10 points]. Consider RV **U** to be discrete with three values -2, 0 and +2 with probability 1/3 each.

(a) **[4 points]** Calculate and plot Conditional Probability Mass Functions, $P_{Y|K}(y|k=+1)$ and $P_{Y|K}(y|k=-1)$. ***Note - clearly mark the exact PMF heights.***

(b) **[2 points]** Calculate and plot the Probability Mass Function, $P_Y(y)$. ***Note - clearly mark the exact PMF heights.***

(c) **[4 points]** Let's now assume that the observed value of the RV **Y** is -1. Mathematically, infer the two probabilities $P_{K|Y}(k|y)$, i.e., find (1) $P_{K|Y}(+1|y=-1)$ and (2) $P_{K|Y}(-1|y=-1)$.

Part 2 [10 points]. In this case, the RV **U** is continuous with a standard Gaussian distribution, i.e., $\mathbf{U} \sim \mathbf{N}(0,1)$.

(a) **[3 points]** Calculate and plot the Conditional PDFs, $f_{Y|K}(y|k=+1)$ and $f_{Y|K}(y|k=-1)$. ***Note - clearly marking the exact PDF heights.***

(b) **[2 points]** Calculate and plot the PDF, $f_Y(y)$.

(c) **[5 points]** Calculate and plot $P_{K|Y}(-1|y)$ for different values of y (i.e., values of observed y are on the x-axis and probabilities $P_{K|Y}(-1|y)$ are on the y-axis). How would you interpret this plot?

See the attached Solution

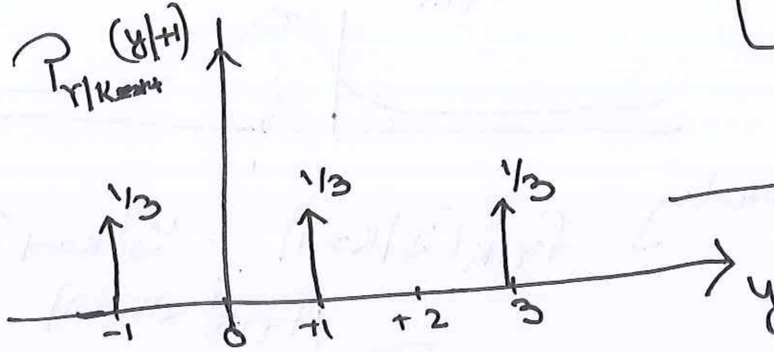
Quiz No 10
↳ Instructor Solution.

Q1) Part 1

(a) $Y = U + K$

$P_{Y|K}(Y|K=+1) \Rightarrow \quad Y = U + 1$

$\Rightarrow Y = -1, +1, +3$ with prob $1/3$ Each. — 1pt

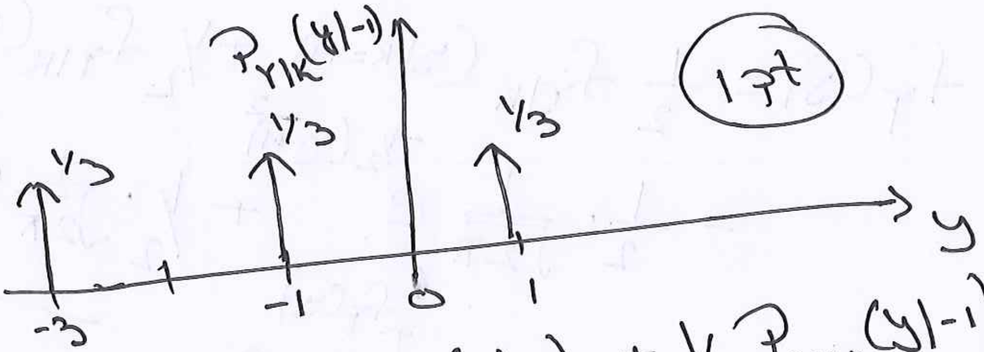


— 1pt

1pt

$P_{Y|K}(y|K=-1) \Rightarrow Y = U - 1$ with prob $1/3$

$\Rightarrow Y = -3, -1, +1$ Each



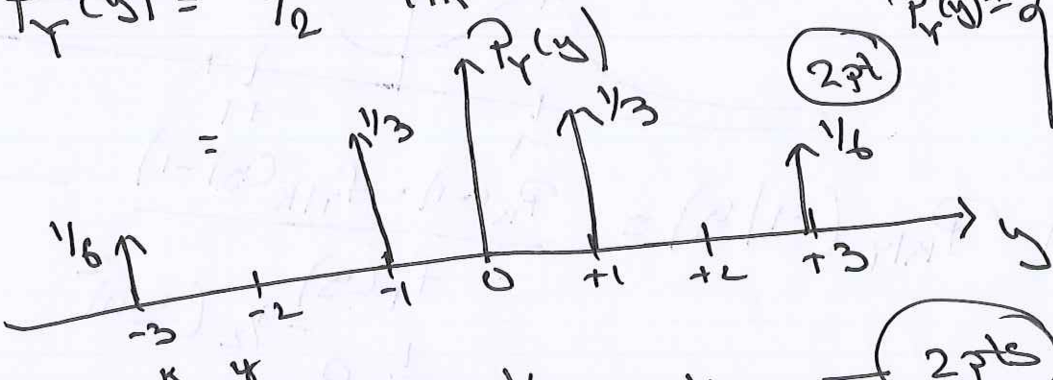
1pt

(b) $P_Y(y) = \frac{1}{2} \cdot P_{Y|K}(y|+1) + \frac{1}{2} P_{Y|K}(y|-1)$

$$P_Y(y) = \begin{cases} -3 - 1/6 \\ -1 - 1/3 \\ +1 - 1/3 \\ +3 - 1/6 \end{cases}$$

(values — Prob)

=



2pt

(c) $P_{K|Y}(+1|-1) = \dfrac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{3}} = \frac{1}{2}$ — 2pts

$P_{K|Y}(-1|-1) = \dfrac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{3}} = \frac{1}{2}$ — 2pts

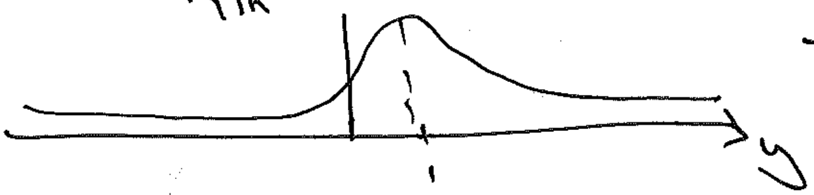
Part 2

(a) $Y = K + U$

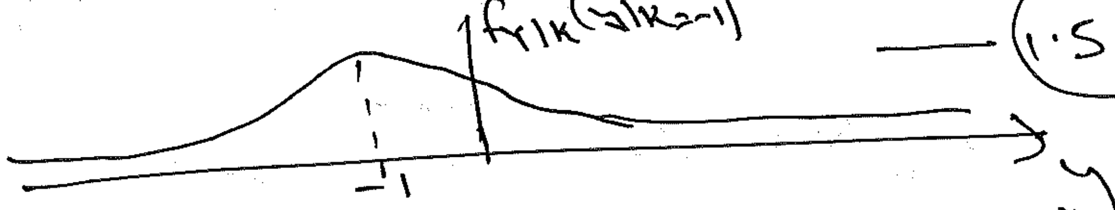
$f_{Y|K}(y|K=+1) \Rightarrow Y = 1 + U$ where $U \sim N(0,1)$

$\Rightarrow Y \approx N(1,1)$

$f_{Y|K}(y|+1)$ given $K=+1$ — 1.5pt

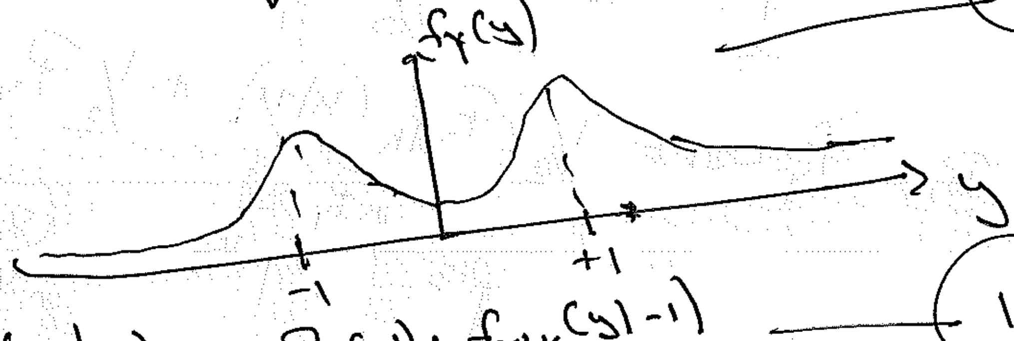


Similarly $f_{Y|K}(y|K=-1)$ $\quad y|K=-1 \sim N(-1,1)$ — 1.5pt



(b) $f_Y(y) = \frac{1}{2} f_{Y|K}(y|K=+1) + \frac{1}{2} f_{Y|K}(y|K=-1)$

$= \frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-1)^2} + \frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y+1)^2}$ — 2pt



— 1pt

(c) $P_{K|Y}(-1|y) = \frac{P_K(-1) \cdot f_{Y|K}(y|-1)}{f_Y(y)}$

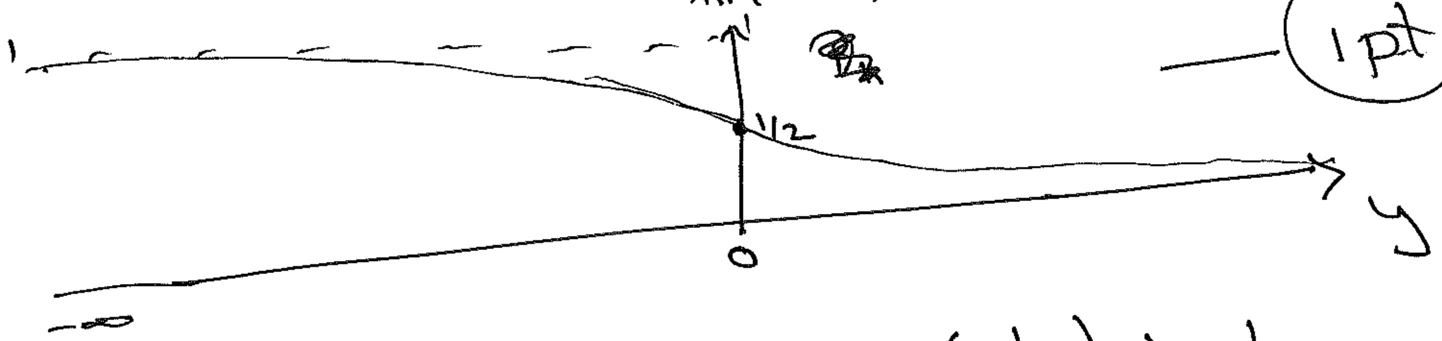
$= \frac{\cancel{\frac{1}{2}} \cancel{\frac{1}{\sqrt{2\pi}}} e^{-\frac{1}{2}(y+1)^2}}{\cancel{\frac{1}{2\sqrt{2\pi}}}\left[e^{-\frac{1}{2}(y-1)^2} + e^{-\frac{1}{2}(y+1)^2} \right]}$

$= \frac{e^{-\frac{1}{2}(y+1)^2}}{\cancel{e^{-\frac{1}{2}(y+1)^2}} \left[\frac{e^{-\frac{1}{2}(y-1)^2}}{e^{-\frac{1}{2}(y+1)^2}} + 1 \right]}$

$$= \frac{1}{1+e^{2y}}$$ 2 pt

$$\Rightarrow P_{K|Y}(-1|y) = \frac{1}{1+e^{2y}}$$

1 pt



for $y \le 0$ $P_{K|Y}(-1|y) \geq 1/2$

& $y > 0$ $P_{K|Y}(-1|y) < 1/2$

So when $y < 0$

$\Rightarrow P_{K|Y}(-1|y)$ is more probable.

1 pt